CATS$^+$

Competishun Advance

Training Sessions

(Lecture - IV)

Que Let $n = 37^{40} - 13^{40}$, then n is divisible by which of the following(s):

(A) 24 ✓ (B) 25 ✓ (C) 26 (D) 27.

$$n = 37^{40} - 13^{40}$$

① Clearly n will be divisible by $37 - 13 = \underline{\underline{24}}$.
(as $x^n - y^n$ will be div. by $x - y$)

② $37^{40} - 13^{40} = \left((37)^2\right)^{20} - \left((13)^2\right)^{20}$ will be div. by $(37)^2 - (13)^2$ as well
$= \underline{(24)(50)}$ ✓

③ Clearly $37^{40} - 13^{40}$ is not div. by 13 hence by 26.

$$27 = 3^3$$

$$37^{40} - 13^{40}$$

$$= (36+1)^{40} - (12+1)^{40}$$

$$= \left({}^{40}C_0 \cdot \underbrace{36}_{9\times 4}{}^{40} + \dots + {}^{40}C_{38} \cdot 36^2 + {}^{40}C_{39} \cdot 36 + 1\right) - \left({}^{40}C_0 \cdot 12^{40} + \dots + {}^{40}C_{38} \cdot 12^2 + {}^{40}C_{39} \cdot 12 + 1\right)$$

$$= \left(27k_1 + 40 \times 36 + \cancel{1}\right) - \left(27k_2 + \frac{\cancel{40}^{\,20} \times \overset{13\times 3}{39}}{\cancel{2}} \cdot 9 \times 16 + 40 \times 12 + \cancel{1}\right)$$

$$= 27k_3 + 40(\underbrace{36-12}_{24 = 8\times 3})$$

$\Rightarrow$ it will not be divisible by 27

$\boxed{\text{Paragraph}}$ Let $(x+2)^{50}$ is divided by $(x-1)^2$, such that remainder is ax^2+bx+c, where $a, b, c \in R$, then answer the followings:-

$\boxed{Q-1}$ Value of a is

(A) 2^{10} (B) 2^9 (C) 0 (D) $50 \cdot 3^{50}$

$\boxed{Q-2}$ If $b = p_1^{\lambda_1} \cdot p_2^{\lambda_2} \cdot p_3^{\lambda_3}$, where p_1, p_2, p_3 are distinct prime numbers then the value of $\lambda_1+\lambda_2+\lambda_3$ is (given $\lambda_1, \lambda_2, \lambda_3 \in N$)

(A) 99 (B) 50 (C) 52 (D) 100

$\boxed{Q-3}$ $|c|$ is

(A) Prime Number (B) greater than 50

(C) Composite prime (D) has 5 divisors.

$$(x+2)^{50} \qquad (x-1)^2$$

$$= \left(x-1+3\right)^{50}$$

$$= {}^{50}C_0 (x-1)^{50} + {}^{50}C_1 (x-1)^{49} \cdot 3 + \cdots + {}^{50}C_{48} (x-1)^2 \cdot 3^{48} + {}^{50}C_{49} (x-1) \cdot 3^{49} + {}^{50}C_{50} \cdot 3^{50}$$

$$= (x-1)^2 \cdot Q_1(x) + 50 \cdot 3^{49} \cdot (x-1) + 3^{50}$$

$$= (x-1)^2 \cdot Q_1(x) + 50 \cdot 3^{49} x - \underbrace{50 \cdot 3^{49} + 3^{50}}$$

$$= (x-1)^2 \cdot Q_1(x) + 50 \cdot 3^{49} \cdot x - 47 \cdot 3^{49}$$

$\lambda_1 = 1, \; \lambda_2 = 49, \; \lambda_3 = 2.$

$= \underset{p_1}{2} \cdot \underset{p_2}{3^{49}} \cdot \underset{p_3}{5^2}$

Remainder is $50 \cdot 3^{49} \cdot x - 47 \cdot 3^{49}$

$ax^2 + bx + c \xrightarrow{\text{Compare}} a = 0, \; b = 50 \cdot 3^{49}, \; c = -47 \cdot 3^{49}.$

Paragraph Let P, Q, R are 3-points on the parabola $y^2=4x$, such that normals at them are concurrent at $N:(15,12)$ then

Answer the followings :-

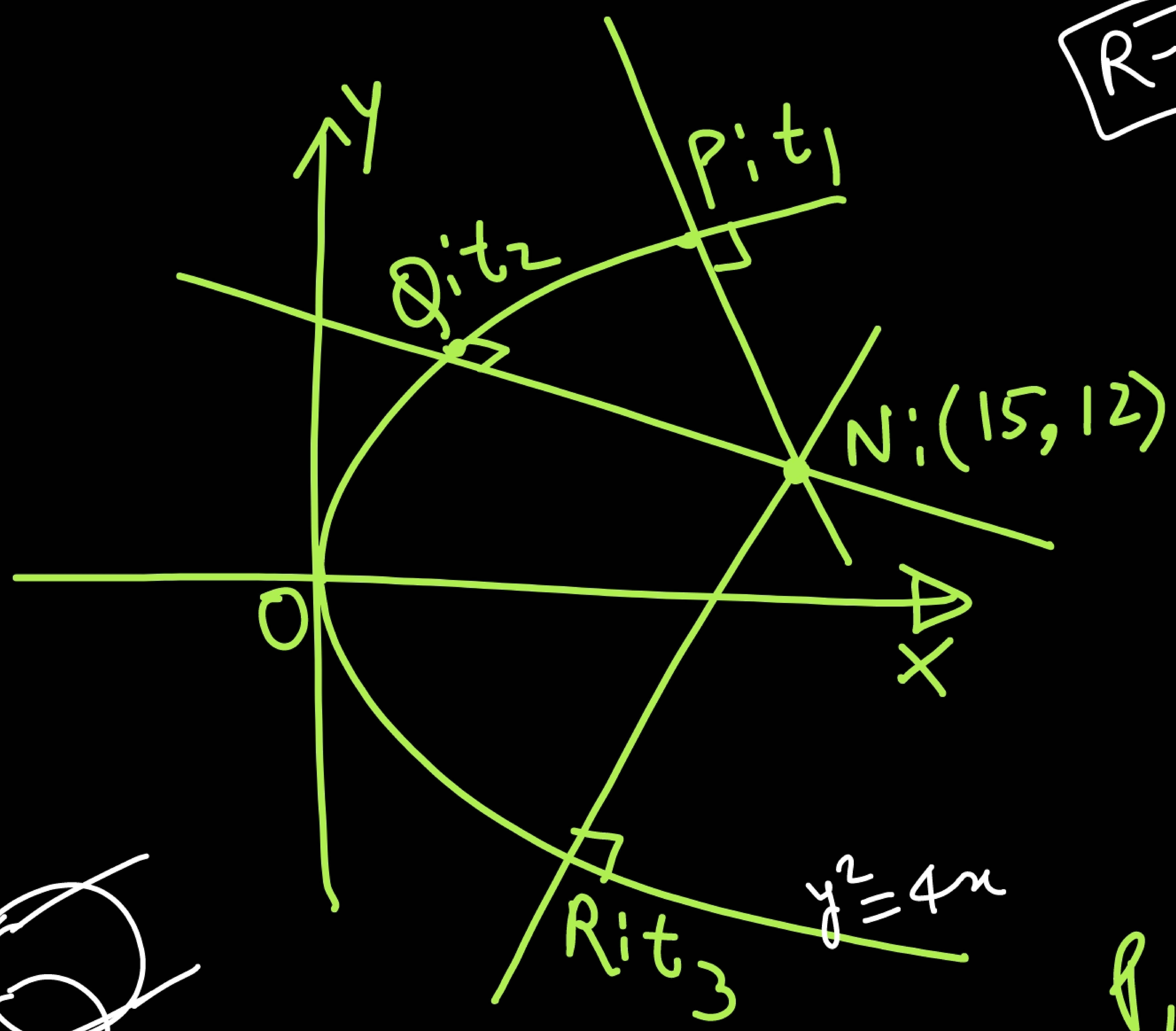
Q-1 If equation of circumcircle of ΔPQR is $x^2+y^2+2gx+2fy+c=0$ then $g+f+c$ is

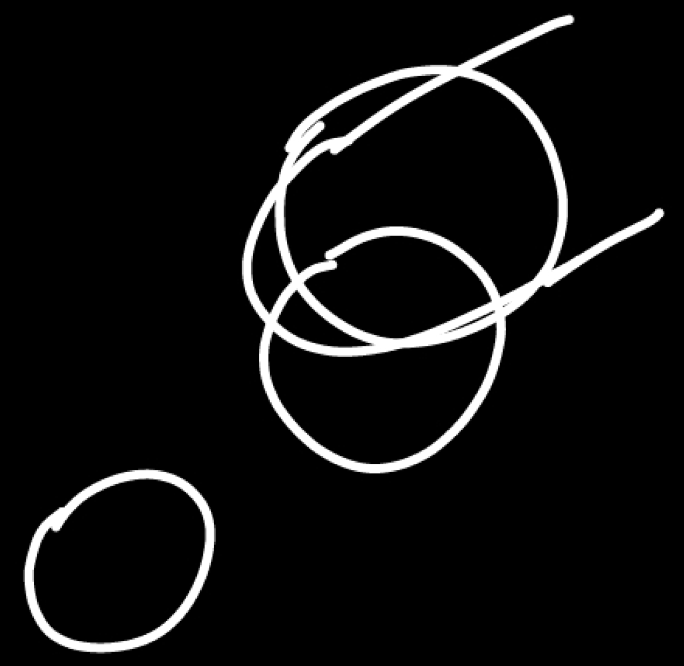
A 0 **B** $\frac{7}{2}$ **C** $\frac{35}{2}$ **D** None of these

Q-2 If coordinates of the centroid of ΔPQR are (α,β) then $|\alpha+\beta|$ is

A 0 **B** 5 **C** $\frac{13}{2}$ **D** None of these

$\left.\begin{array}{l}\alpha=\frac{26}{3}\\ \beta=0\end{array}\right\} \to |\alpha+\beta| = \frac{26}{3}.$





$\boxed{\text{R-1}}$ If P, Q, R are co-normal pts then

$$t_P+t_Q+t_R=0.$$

$\boxed{\text{R-2}}$ If A, B, C, D are 4-concyclic pts on parabola $y^2=4ax$ then

$$t_A+t_B+t_C+t_D=0.$$

<u>Hence</u> If we draw a circle through P, Q, R then it will also cut parabola at vertex.

Eq of normal

$y = -tx + at^3 + 2at$

$\downarrow (15, 12)$

$12 = -15t + t^3 + 2t$

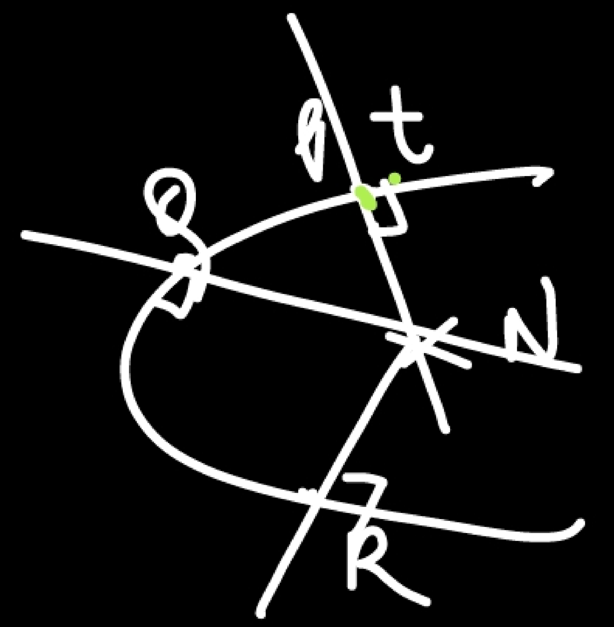
$t^3 - 13t - 12 = 0$ — ①

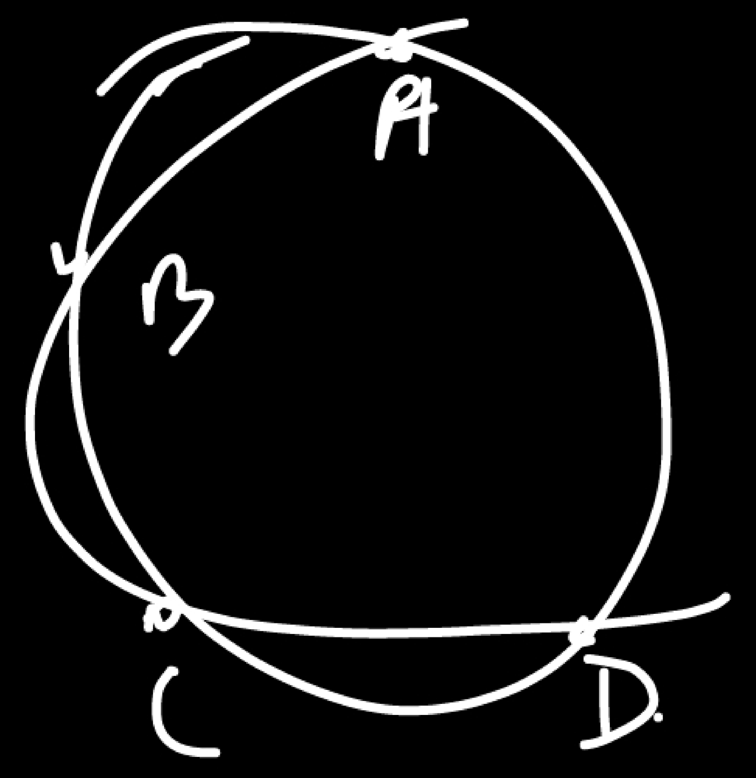
$\hookrightarrow t_1, t_2, t_3$

$t_1 + t_2 + t_3 = 0$

$t_1t_2 + t_2t_3 + t_3t_1 = -13$

$t_1t_2t_3 = 12$.





$x^2 + y^2 + 2gx + 2fy + c = 0$.

$\downarrow (at^2, 2at) = (t^2, 2t)$

$t^4 + 4t^2 + 2gt^2 + 4ft + c = 0$.

$t^4 + (4 + 2g)t^2 + 4ft + c = 0$.

$\hookrightarrow t_A + t_B + t_C + t_D = 0$.

If it is the circle which passes thru P, Q, R

then $t_P + t_Q + t_R + t_K = 0$ when K is 4th pt of intersection

$\hookrightarrow t_K = 0$

$\Rightarrow$ 4th pt is vertex.

Now $t_A \cdot t_B t_C \cancel{t_D} = c$, $t_D = 0$

$\Rightarrow c = 0$.

$$t^4 + (4+2g)t^2 + 4ft = 0 \longrightarrow t_P, t_Q, t_R, t_{Vertex}$$

$$t_{Vertex} = 0$$

$\div t$

$$\Rightarrow t^3 + (4+2g)t + 4f = 0 \longrightarrow ②$$

$$\longrightarrow t_P, t_Q, t_R.$$

$$t^3 - 13t - 12 = 0 \longrightarrow ①$$

$$\longrightarrow t_P, t_Q, t_R$$

As $① \equiv ②$

$$4 + 2g = -13 \Rightarrow g = -\frac{17}{2}$$

$$4f = -12 \Rightarrow f = -3$$

$$\longrightarrow g + f + c = -\frac{17}{2} - 3 + 0 = -\frac{23}{2} \text{ Ans}$$

$$\boxed{x^2 + y^2 - 17x - 6y = 0}$$

$$t^3 - 13t - 12 = 0 \longrightarrow t_1, t_2, t_3$$

$$\sum t_1 = 0$$

$$\sum t_1 t_2 = -13$$

$$\prod t_1 = 12$$

$$\xrightarrow{(\ )^2} \sum t_1^2 + 2\sum t_1 t_2 = 0.$$

$$\sum t_1^2 = 26.$$

$P : (t_1^2, 2t_1)$

$Q : (t_2^2, 2t_2)$

$R : (t_3^2, 2t_3)$

$$G : \left(\frac{\sum t_1^2}{3}, \cancel{\frac{2\sum t_1}{3}} \right) 0$$

$$= \left(\frac{26}{3}, 0 \right)$$

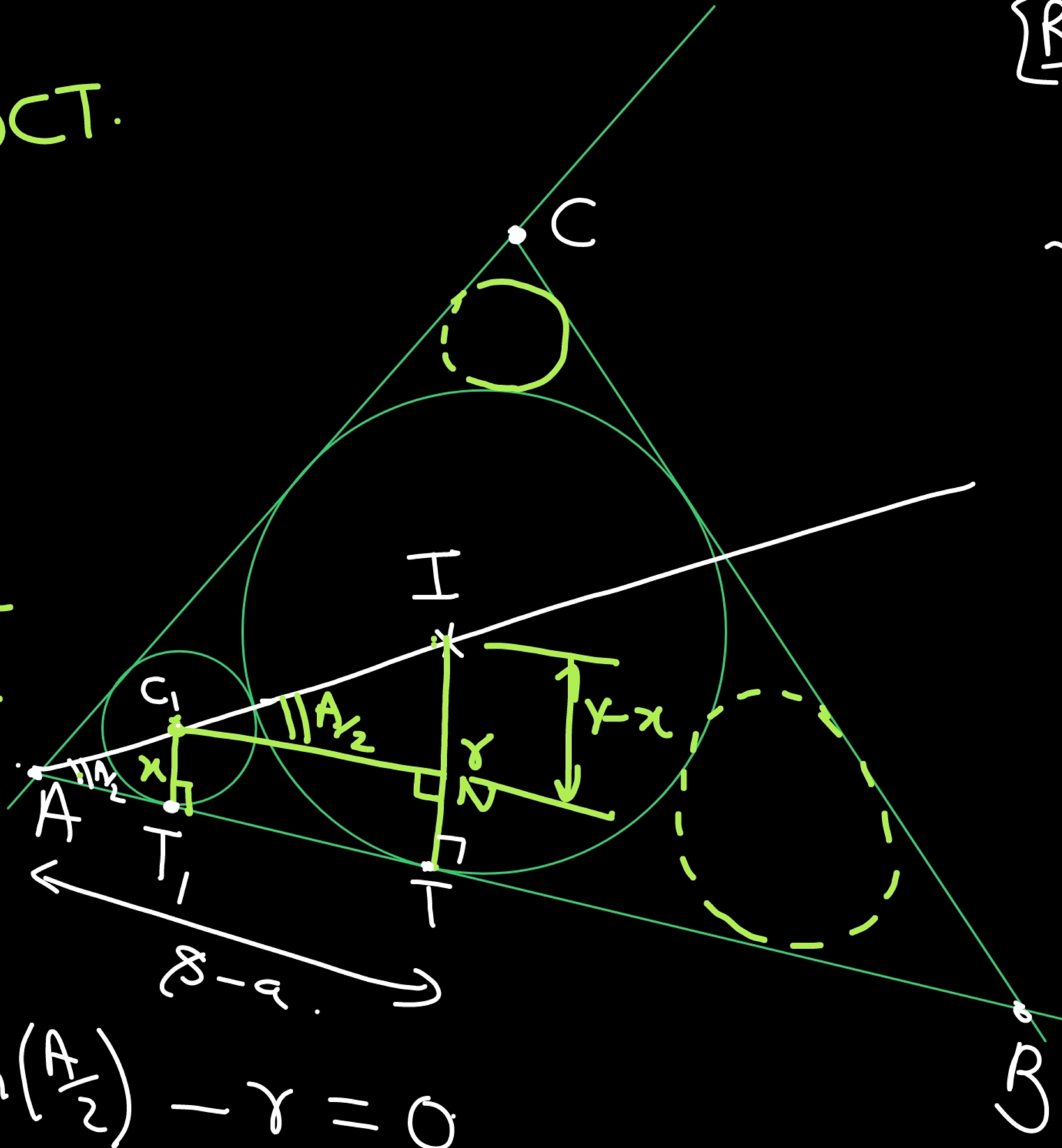
Que Let ABC is a triangle such that $a = 5$, $b = 6$ & $c = 7$. If 3-circles of radii x, y, z are drawn inside ΔABC such that 1st circle touches side AB & BC and also incircle of ΔABC, similarly other two circles are drawn, Now if r is inradius of ΔABC then the value of

$$\frac{\sum \sqrt{xy}}{r}$$ is :-

Ans $\boxed{1}$

$T_1 T$ = length of DCT.

$= 2\sqrt{rx}$

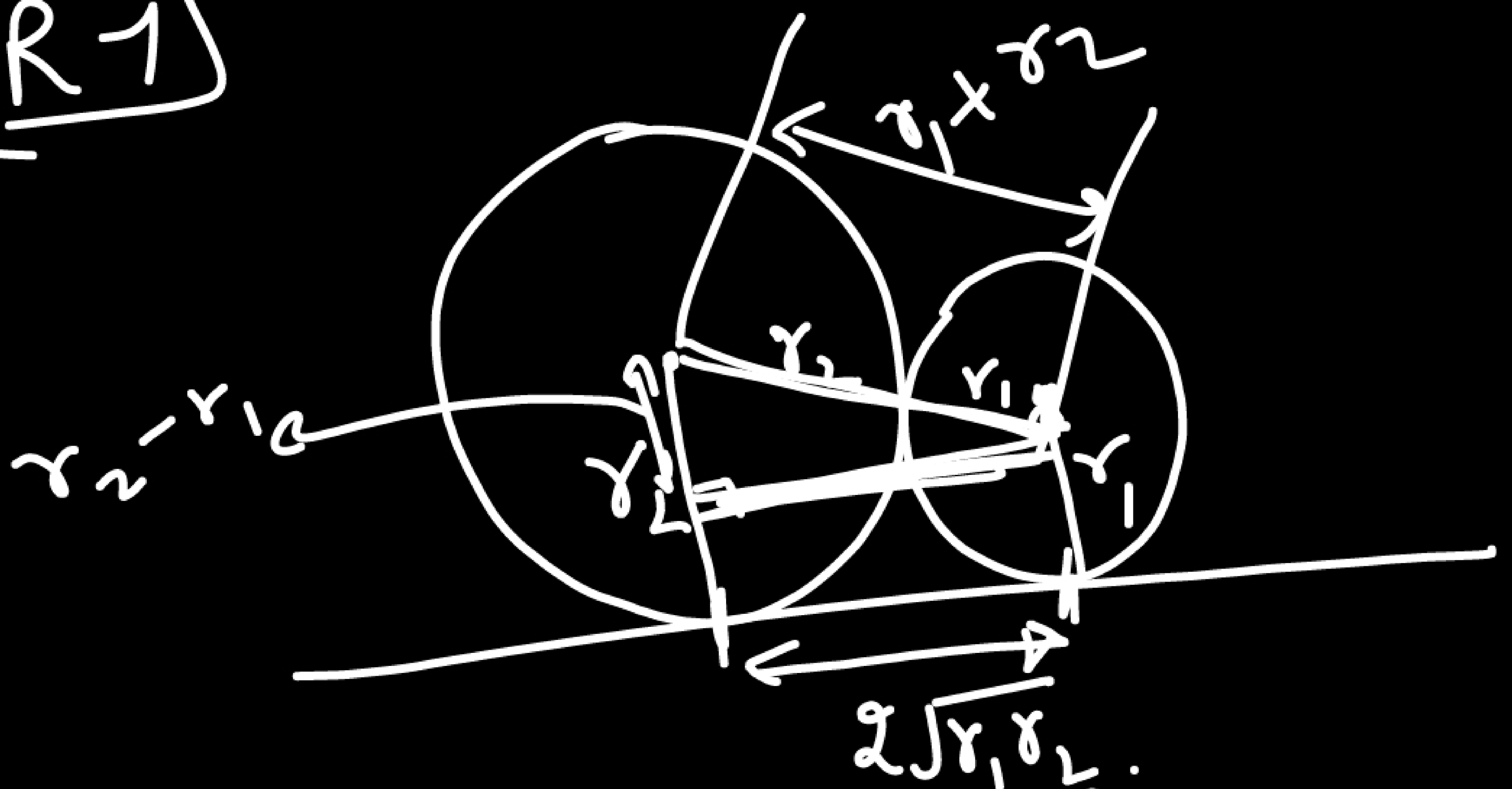


In $\Delta C_1 I N$

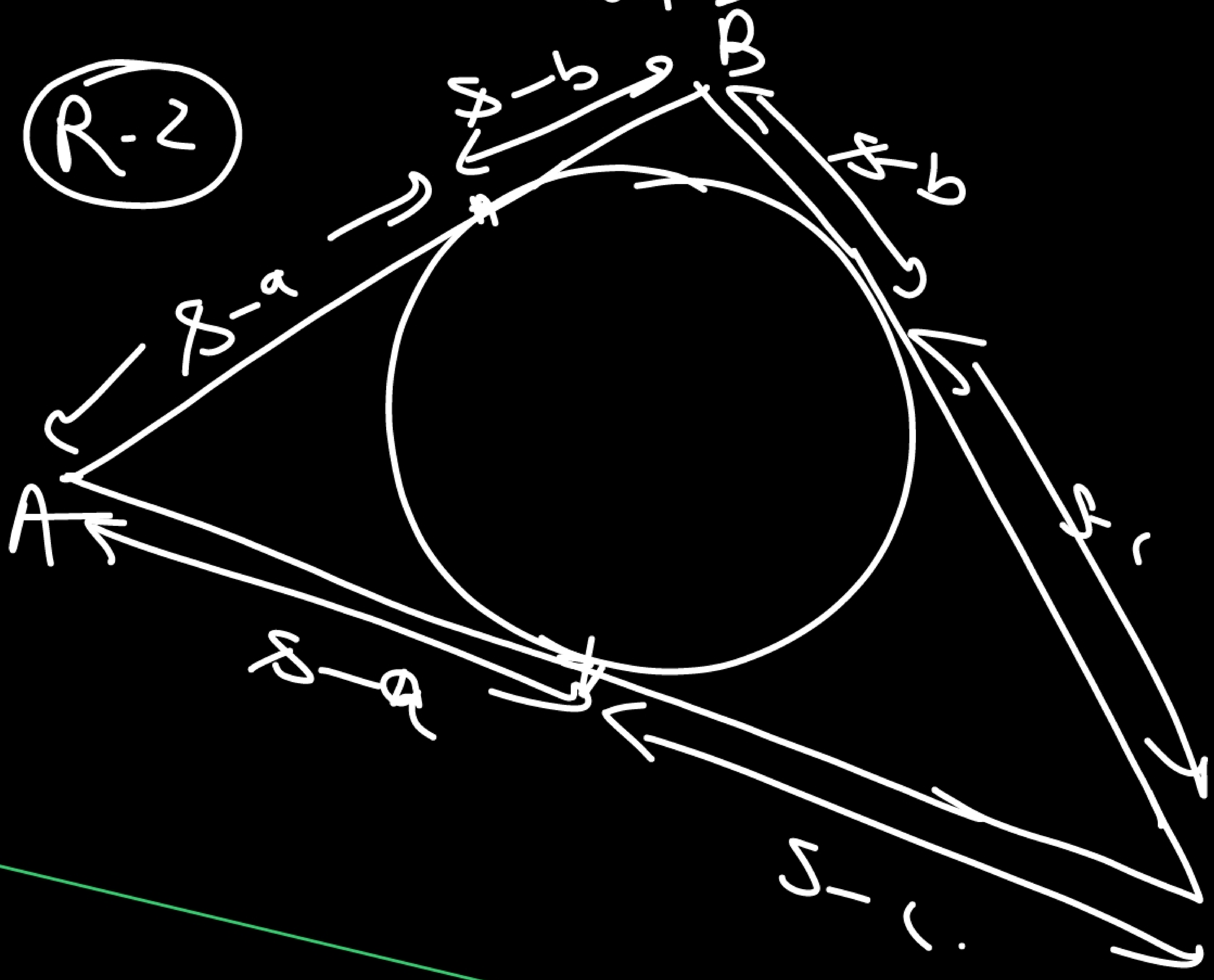
$$\tan\left(\frac{A}{2}\right) = \frac{r-x}{2\sqrt{rx}}$$

$$\Rightarrow x + 2\sqrt{rx}\tan\left(\frac{A}{2}\right) - r = 0$$

R-1



R-2



$$x + \left(2\sqrt{r}\tan\left(\frac{A}{2}\right)\right)\sqrt{x} - r = 0.$$

$$\sqrt{x} = \frac{-\cancel{2}\sqrt{r}\tan\left(\frac{A}{2}\right) \pm \sqrt{\cancel{4}r\tan^2\left(\frac{A}{2}\right) + \cancel{4}r}}{\cancel{2}}$$

$$\sqrt{x} = \sqrt{r}\left(-\tan\left(\frac{A}{2}\right) \pm \sqrt{\tan^2\left(\frac{A}{2}\right)+1}\right) = \sqrt{r}\left(-\tan\left(\frac{A}{2}\right) \pm \sec\left(\frac{A}{2}\right)\right)$$

$$\sqrt{x} = \sqrt{r}\left(\sec\left(\frac{A}{2}\right) - \tan\left(\frac{A}{2}\right)\right)$$

$$\boxed{\frac{\sqrt{x}}{\sqrt{r}} = \frac{1-\sin\left(\frac{A}{2}\right)}{\cos\left(\frac{A}{2}\right)}} \Rightarrow$$

similarly

$$\frac{\sqrt{y}}{\sqrt{r}} = \frac{1-\sin\left(\frac{B}{2}\right)}{\cos\left(\frac{B}{2}\right)}$$

$$\therefore \frac{\sqrt{xy}}{r} = \frac{\left(1-\sin\left(\frac{A}{2}\right)\right)\left(1-\sin\left(\frac{B}{2}\right)\right)}{\cos\left(\frac{A}{2}\right)\cos\left(\frac{B}{2}\right)}$$

$$\frac{\sum\sqrt{xy}}{r} = \sum \frac{\sqrt{xy}}{r} = \sum \frac{\left(1-\sin\left(\frac{A}{2}\right)\right)\cdot\left(1-\sin\left(\frac{B}{2}\right)\right)}{\cos\left(\frac{A}{2}\right)\cos\left(\frac{B}{2}\right)}$$

$$= \frac{\sum\left(1-\sin\left(\frac{A}{2}\right)\right)\left(1-\sin\left(\frac{B}{2}\right)\right)\cdot\cos\left(\frac{C}{2}\right)}{\cos\left(\frac{A}{2}\right)\cdot\cos\left(\frac{B}{2}\right)\cdot\cos\left(\frac{C}{2}\right)} = \frac{\sum\left(1-s\left(\frac{A}{2}\right)-s\left(\frac{B}{2}\right)+s\left(\frac{A}{2}\right)s\left(\frac{B}{2}\right)\right)c\left(\frac{C}{2}\right)}{c_1c_2c_3}$$

$$= \frac{\sum\left(c_3 - s_1c_3 - s_2c_3 + s_1s_2c_3\right)}{c_1c_2c_3}$$

$$= \frac{\sum c_1 - \sum\left(s_1c_3+s_3c_1\right) + \sum s_1s_2c_3}{c_1c_2c_3} = \frac{\cancel{\sum c_1 - \sum \overbrace{\sin\left(\frac{A}{2}+\frac{C}{2}\right)}^{\cos\left(\frac{B}{2}\right)}} + \sum s_1s_2c_3}{\prod c_1}$$

$$= \frac{\sum s_1s_2c_3}{\prod c_1} = \frac{\prod c_1}{\prod c_1} = \boxed{1}$$

for this P.T.O.

$$\cos(\alpha+\beta+\gamma) = \cos(\alpha)\cdot\cos(\beta)\cos(\gamma) - \sum \sin(\alpha)\sin(\beta)\cdot\cos(\gamma)$$

$$\Big\downarrow \; \alpha = A/2, \quad \beta = B/2, \quad \gamma = C/2$$

$$\cancelto{0}{\cos\left(\frac{A+B+C}{2}\right)} = \prod \cos\left(\frac{A}{2}\right) - \sum \sin\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)\cos\left(\frac{C}{2}\right)$$

$$\hookrightarrow \sum \sin\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)\cos\left(\frac{C}{2}\right) = \prod \cos\left(\frac{A}{2}\right)$$